GIS-Based Enhanced Farm Viability Assessment

Using Multi-Source Remote Sensing and Soil Data

Smallholder Farmer Onboarding — Kachia LGA, Kaduna State, Nigeria

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1. Introduction

ThriveAgric is a fast-growing agricultural technology company dedicated to ensuring food security while building the largest network of profitable smallholder farmers across Africa. One of ThriveAgric's operational model is Farm Mapping; the systematic identification, measurement, and viability assessment of farmlands, this is done prior to input financing and loan disbursement. This process ensures that agricultural loans are disbursed toward farms with the highest potential for yield and repayment.

This project presents an enhanced GIS-based Farm Viability Assessment for Kachia Local Government Area (LGA), Kaduna State, one of Nigeria's leading ginger producing zones. Using high-resolution soil property data from iSDAsoil (30m), moisture analysis derived from Landsat 8 NDMI via Google Earth Engine, and a Soil Quality Index (SQI) combining organic carbon, nitrogen, phosphorus and pH, 24 sample farms were assessed across eight environmental factors to generate a Farm Viability Score for each farm.

Kachia LGA is widely known as a major ginger producing zone in Kaduna State. The soil conditions identified in this study; slightly acidic pH (5.5–6.4) and Sandy Loam/Clay Loam textures, are highly consistent with optimal ginger cultivation requirements, further validating the agricultural potential of the study area and the relevance of ThriveAgric's input financing programme in this region.

2. Objectives

- Assess the agricultural viability of 24 sample smallholder farms in Kachia LGA using GIS-based spatial analysis
- Generate a composite Farm Viability Score using eight environmental factors at 30m resolution
- Classify farms using both FAO international land suitability standards and relative local prioritization
- Derive a Soil Quality Index (SQI) as a composite measure of soil fertility status
- Produce professional cartographic outputs aligned with ThriveAgric's farm mapping workflow

3. Study Area

Kachia LGA is located in the southern part of Kaduna State, Nigeria. The LGA is predominantly rural with smallholder farming as the primary livelihood activity. It is renowned as a global hub for ginger production. Other crops grown in the area include maize, sorghum, groundnuts, and these are all crops supported under ThriveAgric's input financing programme.

The area is characterised by gently undulating terrain with some steep ridges in the central zone, a network of rivers and seasonal waterbodies, and soils dominated by Sandy Loam and Clay Loam textures. Land cover is predominantly cropland and grassland, indicating significant existing agricultural activity across the LGA.

4. Data Sources

Dataset	Source	Description	Resolution
Digital Elevation Model	USGS EarthExplorer	SRTM DEM for slope derivation	30m
Organic Carbon	iSDAsoil	Soil organic carbon 0–20cm depth	30m
Total Nitrogen	iSDAsoil	Total nitrogen 0–20cm depth	30m
Phosphorus	iSDAsoil	Extractable phosphorus 0–20cm depth	30m
Soil pH	iSDAsoil	pH water 0–20cm depth	30m
Soil Texture	iSDAsoil	USDA texture classification 0–20cm	30m
NDMI Moisture	GEE Landsat 8	Normalized Difference Moisture Index May 2024	30m
Rivers	OpenStreetMap	River polylines for Kachia LGA	Vector
Waterbodies	OpenStreetMap	Dam and waterbody polygons	Vector
Farm Polygons	Google Earth Pro	24 digitized sample farm polygons	Vector

5. Methodology

All spatial data was processed in QGIS 3.40 (Bratislava) and reprojected to WGS 1984 UTM Zone 32N (EPSG:32632). Soil property data was sourced from iSDAsoil at 30m resolution, specifically designed for African smallholder farming contexts. Landsat 8 imagery was processed via Google Earth Engine using a cloud-filtered mosaic of 13 scenes from May to July 2024.

5.1 Data Preparation

- All datasets were clipped to Kachia LGA administrative boundary
- All layers were reprojected to EPSG:32632 (WGS 84 UTM Zone 32N)
- iSDAsoil layers back-transformed using published scale factors: SOC and Phosphorus using exp(x/10)-1, Nitrogen using exp(x/100)-1, pH using x/10
- 24 sample farm polygons digitized in Google Earth Pro and imported as KML
- River and waterbody layers rasterized at 30m then processed with GDAL Proximity tool

5.2 Suitability Factor Analysis

Eight environmental factors were reclassified to a 1–5 suitability scale and combined using weighted overlay:

Factor	Source	Weight	Justification
Organic Carbon	iSDAsoil 30m	20%	Primary soil fertility indicator
NDMI Moisture	GEE Landsat 8	15%	Direct moisture proxy during planting season
Total Nitrogen	iSDAsoil 30m	15%	Key macronutrient for crop yield
Soil Texture	iSDAsoil 30m	15%	Physical soil structure and drainage
Soil pH	iSDAsoil 30m	10%	Nutrient availability gateway
Slope	USGS SRTM	10%	Terrain constraint for mechanisation
Water Distance	OpenStreetMap	10%	Proximity to irrigation source
Phosphorus	iSDAsoil 30m	5%	Secondary macronutrient

5.3 Proximity Analysis

River and waterbody layers were rasterized separately at 30m resolution. GDAL Proximity was applied to each to generate Euclidean distance surfaces. The two rasters were combined using Raster Calculator minimum () to produce a single water proximity raster representing distance to nearest water source.

5.4 Weighted Overlay

The eight reclassified suitability rasters were combined in QGIS Raster Calculator:

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Viability = (Carbon×0.20) + (NDMI×0.15) + (Nitrogen×0.15) + (Texture×0.15)
      + (pH×0.10) + (Slope×0.10) + (Water×0.10) + (Phosphorus×0.05)
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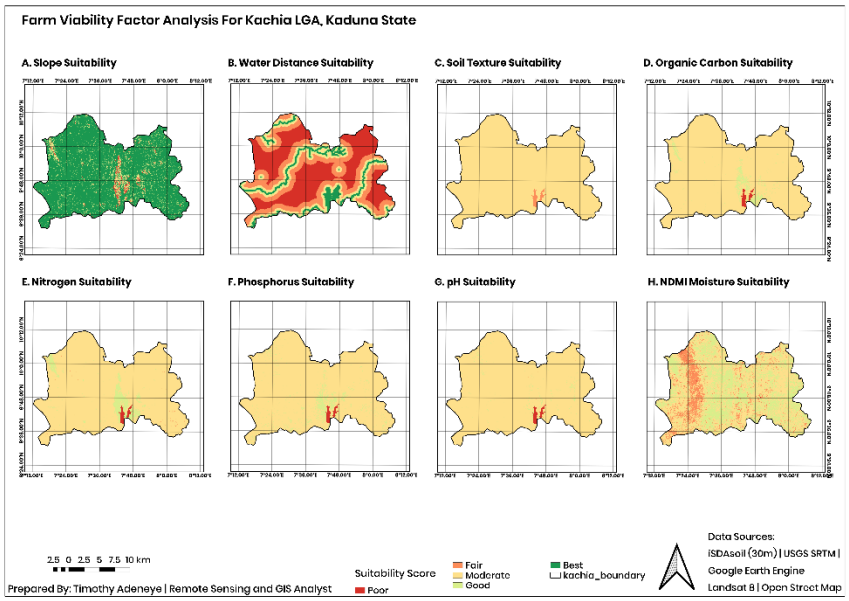

5.5 Farm Scoring and Classification

Zonal Statistics extracted mean scores per farm from all factor rasters and the composite viability raster. Farms were classified using two complementary frameworks:

Framework	Classification	Score Range	Meaning
FAO Absolute	S1 — Highly Suitable	≥ 4.0	No significant limitations
FAO Absolute	S2 — Moderately Suitable	3.0 – 3.99	Minor manageable limitations
FAO Absolute	S3 — Marginally Suitable	2.0 – 2.99	Significant limitations
Relative Priority	High	≥ 3.25	Best farms in Kachia; prioritize for loans
Relative Priority	Medium	3.10 – 3.25	Suitable with standard conditions
Relative Priority	Low	< 3.10	Requires field verification

5.6 Soil Quality Index (SQI)

A Soil Quality Index was derived by normalizing four soil properties to a 0–1 scale using min-max normalization and combining with weighted aggregation:

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SQI = (Carbon×0.30) + (Nitrogen×0.25) + (Phosphorus×0.25) + (pH×0.20)
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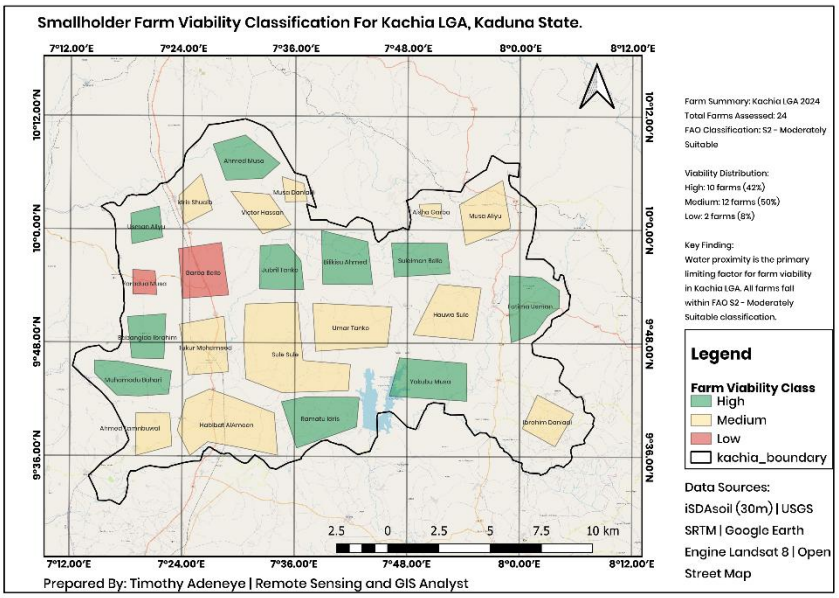
The SQI provides a composite measure of soil fertility status on a 0–1 scale, directly indicating which farms require the most intensive input support from ThriveAgric's financing programme.

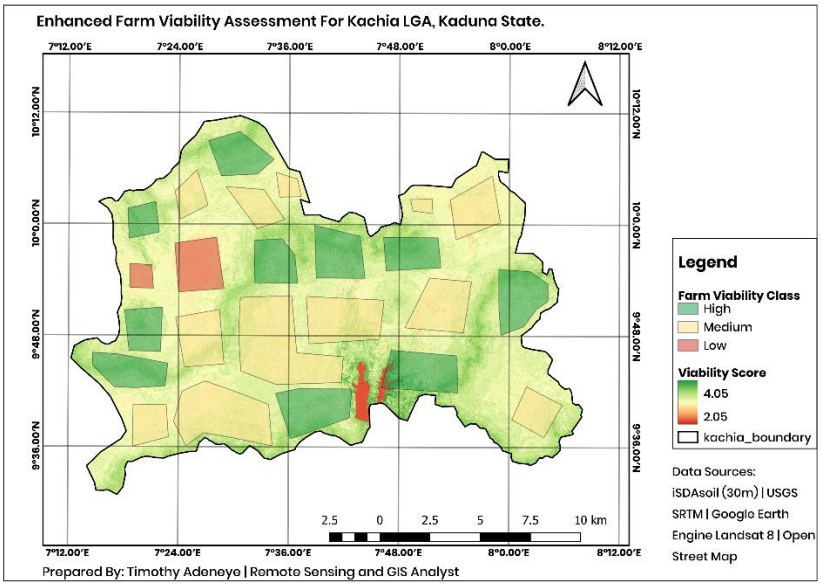
6. Results and Findings

6.1 Farm Viability Distribution

Farm Viability Scores ranged from 3.00 to 3.35 across 24 sampled farms. All farms fall within the FAO S2 — Moderately Suitable classification, reflecting Kachia's agricultural potential alongside manageable environmental constraints.

Viability	Farms	%	FAO Class	Loan Recommendation
High	10	42%	S2	Immediate disbursement recommended
Medium	12	50%	S2	Disbursement with standard conditions
Low	2	8%	S2	Field verification required first

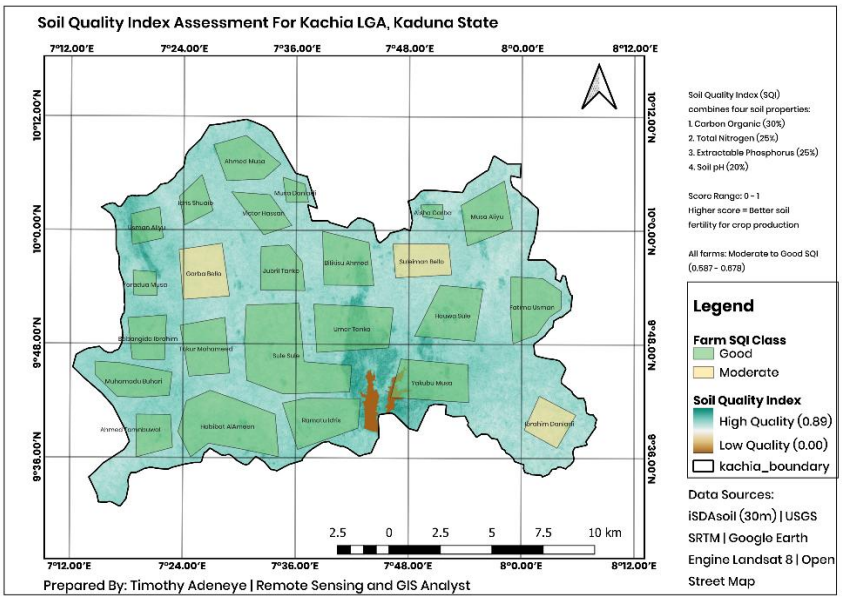




6.2 Soil Quality Index Results

SQI values ranged from 0.587 to 0.678 across all 24 farms, placing all farms in the Moderate to Good soil quality category. This confirms that Kachia soils have adequate baseline fertility that responds positively to targeted fertilizer inputs.

SQI Class	SQI Range	Farms	Implication
Good	0.60 – 0.678	~18 farms	Soil responds well to standard inputs
Moderate	0.587 – 0.60	~6 farms	Requires enhanced input package



6.3 Key Findings

- All 24 farms fall within FAO S2; Moderately Suitable, confirming Kachia's agricultural potential with manageable limitations.
- Water proximity is the primary limiting factor; most rivers in Kachia are seasonal, creating moisture stress during dry season and directly impacting loan repayment risk.
- Slope is not a significant constraint; Kachia is predominantly flat (score 4–5), ideal for mechanised smallholder farming.
- Soil organic carbon (0.11–12.46 g/kg) and nitrogen (0.01–1.56 g/kg) are moderate, confirming soils will respond well to ThriveAgric's fertilizer input programme.
- SQI ranges from 0.587–0.678 (Moderate to Good); consistent soil quality across all farms with room for improvement through targeted inputs.
- Kachia's slightly acidic pH (5.5–6.4) and Sandy Loam/Clay Loam textures are ideal for ginger, maize and sorghum; Kachia's primary crops.
- 42% of farms are High Priority for immediate loan disbursement, representing strong near term portfolio opportunity for ThriveAgric.

7. Recommendations

- Prioritise the 10 High viability farms for immediate input loan disbursement, these have the highest potential return-on-investment within Kachia LGA.
- Implement targeted irrigation support for Low viability farms in water-scarce zones, proximity to water is the single most impactful improvement lever.
- Use SQI scores to calibrate fertilizer input packages, farms with lower SQI scores require enhanced nitrogen and phosphorus inputs before planting.
- Conduct field verification for the 2 Low viability farms before disbursing loans, on-ground soil testing may reveal localised conditions not captured in satellite data.
- Expand this viability model to additional LGAs in Kaduna State and other locations to support ThriveAgric's regional onboarding and expansion strategy.

8. Tools and Technologies

Tool	Purpose
QGIS 3.40 (Bratislava)	Primary GIS platform; data processing, spatial analysis, map production
Google Earth Engine	Landsat 8 mosaic processing, NDMI calculation, cloud filtering
iSDAsoil (30m)	High-resolution African soil property data; carbon, nitrogen, phosphorus, pH, texture
GDAL Proximity	Euclidean distance raster generation for river and waterbody layers
GDAL Raster Calculator	Scale factor back-transformation and SQI normalization
QGIS Raster Calculator	Weighted overlay computation
QGIS Zonal Statistics	Farm-level score extraction from raster layers
Google Earth Pro	Digitizing of 24 sample farm polygons across Kachia LGA

9. Limitations

- Farm polygons are simulated from Google Earth Pro imagery and do not represent live farms. They are hypothetical just for the purpose of the analysis and results should be validated with surveyed farm boundaries.
- iSDAsoil data at 30m is modelled from field samples and may not capture highly localised soil variability at the individual farm level.
- NDMI from Landsat 8 is a proxy for moisture availability, it reflects vegetation water content rather than direct soil moisture measurement.
- River data from OSM does not distinguish between perennial and seasonal watercourses, dry season water access may be overestimated for some farms.
- Rainfall and temperature data were not incorporated. Future iterations should include climatic suitability factors for a more comprehensive assessment

10. Conclusion

This enhanced GIS-based Farm Viability Assessment demonstrates how multi-source remote sensing and high-resolution soil data can directly support ThriveAgric's farm onboarding and loan disbursement workflow. By combining eight environmental factors, including iSDAsoil soil properties at 30m resolution and Landsat 8 moisture analysis via Google Earth Engine, into a single composite viability score, field officers and credit analysts can make faster, more evidence-based financing decisions.

The finding that all 24 sampled farms in Kachia LGA fall within FAO S2 - Moderately Suitable classification confirms the area's agricultural potential. The 42% of farms achieving High viability represent an immediate loan disbursement opportunity, while the identification of water proximity as the primary limiting factor provides a clear, actionable basis for targeted irrigation investment to unlock the remaining farms.

Kachia's status as a leading ginger producing zone in Kaduna State, combined with its slightly acidic soils and moderate fertility levels that respond well to fertilizer inputs, makes it an ideal operational zone for ThriveAgric's model.

This methodology is scalable and replicable across several states in Nigeria and beyond, providing a robust and cost-effective framework for evidence-based agricultural financing at scale.